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Integrating Spatial Autocorrelation Analysis Using Ambient Noise and Gravity Measurements for Subsurface Characterization

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Abstract

This study investigates the application of SPatial AutoCorrelation (SPAC) analysis using seismic noise and gravity surveying as a novel geophysical workflow for characterizing subsurface structure of urban areas. The primary objective is to determine subsurface's velocity through seismic noise analysis and, subsequently, estimate their density values using well established empirical relationships. The calculated density values will serve as inputs to refine gravity measurements and enhance subsurface modeling for the investigated area. The proposed methodology was applied to King Fahd University of Petroleum and Minerals (KFUPM) campus in Dhahran, Saudi Arabia. Seismic noise was recorded using a network of broadband wireless sensors spatially distributed within the campus. The analysis of seismic noise recorded at different aperture arrays allowed the determination of phase velocities in a wide frequency range. By inverting this information, the shear-wave velocities of soils (Vs) were extracted at various depths. Exploiting empirical relationships proposed in literature for similar geological environments, the 1D shear-wave velocity profiles were converted to density estimates with depth. These values served as effective inputs to review gravity results available in KFUPM campus and enhance the accuracy of the proposed 2D gravity models. Moreover, the velocity models derived from the SPAC method can be used to constrain and refine the number of subsurface layers interpreted from the gravity data. The derived shear wave velocity and density profiles revealed subsurface structures with distinct seismic properties corresponding to sedimentary sequences and possible tectonic features beneath the campus. The results revealed an ENE-WSW trend contrasting with the primary NNW-SSE joint set observed on the campus. This ENE-WSW orientation aligns with the horizontal stress axis σ_H, which related to regional intraplate stresses from the Arabian-Eurasian collision, highlighting complex interactions between regional and local tectonic forces. The result of the study makes the combination of SPAC analysis with gravity measurements a promising approach to investigate complex geological environments in urban areas.

Introduction

Urban geophysical investigations have increasingly become vital in addressing the challenges of subsurface characterization in densely populated and infrastructure-heavy environments (Gabàs et al., 2014). As cities expand and institutions grow, understanding the underlying geologic structures becomes increasingly critical for hazard assessment, infrastructure planning, groundwater resource management, and geotechnical design. In this context, passive seismic methods have gained prominence due to their non-invasive nature, cost-effectiveness, and applicability in urban settings with minimal disruption to surface activity (Cheng et al., 2015; Morgan et al., 2022). This study focuses on the application of the SPatial AutoCorrelation analysis of seismic noise to investigate Vs profiles underneath the King Fahd University of Petroleum and Minerals (KFUPM) campus. The ultimate aim is to enhance gravity-based subsurface modeling by incorporating SPAC-derived velocity and density constraints, thereby improving geological interpretation and tectonic understanding in the region. The Eastern Province of Saudi Arabia, where the KFUPM campus is located, holds significant geologic interest due to its position on the crest of the Dammam Dome. This structure, associated with hydrocarbon entrapment and regional tectonic evolution (Weijermars, 1999), represents a critical site for understanding the interaction between sedimentary sequences and intraplate stress regimes driven by the distant Arabian-Eurasian plate convergence. Historically, the subsurface configuration of the Dammam Dome has been explored using gravity methods due to their sensitivity to density contrasts and their suitability in areas where seismic surveys are limited by resolution or logistical constraints. Recent studies, including Chavanidis et al. (2022), have detailed the implementation of a high-resolution gravity survey across the KFUPM campus, revealing subvertical density anomalies with ENE-WSW orientations that align with regional tectonic stress patterns. However, gravity data interpretation inherently suffers from non-uniqueness, and additional constraints such as density or velocity profiles are essential to refine models and reduce ambiguity.

To address this, the present study integrates seismic-derived shear wave velocities with gravity modeling by leveraging the SPAC method. SPAC analysis provides an effective method for retrieving phase velocity dispersion curves from seismic noise, which can be inverted to estimate Vs profiles at depth (Cho & Iwata, 2020; Luo et al., 2016). These profiles not only provide insights into the mechanical behavior of subsurface layers but also allow the conversion of shear velocities to density values using established empirical relationships (e.g. Gardner et al., 1974; Brocher, 2005). By deploying a network of broadband wireless seismic sensors across the campus, this study leverages urban ambient noise fields to map subsurface stratigraphy and refine gravity models with enhanced accuracy. One of the key advantages of SPAC over traditional seismic approaches is its adaptability to urban settings. In contrast to controlled-source seismic surveys, which often require significant energy input and logistical coordination, SPAC utilizes naturally occurring microtremors, thereby circumventing issues related to active source generation in noise-sensitive or restricted areas. Moreover, SPAC's ability to operate over a range of aperture sizes and frequencies enables its application in delineating both shallow and moderately deep structures, depending on array geometry and ambient noise conditions.

At KFUPM, the urban campus environment poses several challenges for geophysical data acquisition, including high anthropogenic noise, variable surface conditions, and spatial constraints for sensor deployment. Therefore, careful planning of sensor array configurations, data duration, and noise filtering protocols was essential. The sensors were arranged in circular arrays with varying radii to capture a broad range of frequencies. The integration of SPAC-derived density estimates into gravity data interpretation provides a powerful synergy. Gravity inversion, while effective in delineating lateral density variations, often lacks vertical resolution and is sensitive to the assumed density model. By anchoring the inversion with real density values derived from seismic data, the model space is reduced, and the resulting gravity models become more geologically plausible. In the case of KFUPM, this approach has yielded refined interpretations of subsurface structures, revealing layers consistent with known sedimentary units such as the Rus and Dammam Formations, as well as features indicative of tectonic deformation.

The relevance of this integrated approach extends beyond the campus boundaries. The observed ENE-WSW trending anomalies from both SPAC and gravity data align with the regional stress field orientation associated with the Arabian-Eurasian collision zone. While previous fracture mapping in the Dammam Dome has emphasized NNW-SSE joint sets, the SPAC and gravity data suggest an underlying stress regime with a more complex geometry. This is supported by recent interpretations of soft-sediment detachments and transtensional stress regimes affecting the area, as documented by Tranos & Osman (2021). Moreover, the use of SPAC in urban geophysical applications is part of a broader trend toward passive, low-impact methodologies that align with environmental and logistical considerations in modern research. As smart cities, campuses, and industrial zones become increasingly data-driven and infrastructure-intensive, non-invasive subsurface monitoring techniques will play a growing role in these areas. In summary, this introduction lays the groundwork for a comprehensive investigation into the use of SPAC analysis for shear wave velocity and density estimation at the KFUPM campus. By combining these results with gravity data, the study addresses key questions related to subsurface structure, tectonic setting, and the methodological integration of geophysical tools in complex urban environments. The following sections will detail the methodologies employed, the processing steps taken to ensure data reliability, the inversion techniques used to extract Vs and density, and the final implications for subsurface modeling and tectonic interpretation in the context of the Dammam Dome.

Data and processing

To characterize the shallow subsurface shear wave velocity (Vs) structure of the KFUPM campus, a passive seismic survey was implemented. The field campaign utilized a network of broadband wireless seismometers arranged in concentric circular arrays at two strategically selected locations within the campus (Fig. 1a). The location of the arrays was chosen to maximize coverage over areas with relatively low urban noise, sufficient open space for array geometry, and geological relevance based on previous gravity surveys (Soupios et al., 2021). Each array configuration (Fig. 1b) included three concentric rings of sensors with increasing aperture radii to enable multi-scale frequency analysis. Noise was recorded simultaneously across all arrays for 4 to 8 hours, depending on the local noise conditions and logistical constraints.

Seismic data acquisition was carried out using ten triaxial IGU-BD3C-5 broadband wireless sensors. Each sensor was programmed through the associated acquisition software with carefully selected parameters optimized for ambient seismic noise analysis. A sampling rate of 0.5 milliseconds was used to ensure high temporal resolution, and the devices operated in continuous mode to enable uninterrupted recording throughout the deployment duration. The anti-aliasing filter was set to linear phase to preserve waveform integrity, while a low-cut filter at 0.2 Hz was applied to suppress low-frequency tilt and environmental noise. Each sensor was recorded with a uniform gain setting of 6 dB to ensure balanced sensitivity across all directions. The sensors were configured with damping ratios between 1.040 and 1.560, indicating an overdamped response. This setup helps minimize oscillations and ensures a smooth return to equilibrium after ground motion, improving signal stability. Additionally, the resonance frequencies of the sensors ranged from 0.16 Hz to 0.24 Hz, meaning they are susceptible to low-frequency vibrations. A built-in GPS synchronized the stations. Station positioning was obtained using a Trimble R10+ Differential GPS.

Given the urban setting of the KFUPM campus, special attention was given to sensor emplacement and noise mitigation. Sensors were deployed directly on exposed ground surfaces (Figure 1c) and covered to reduce wind and temperature effects. Where necessary, sensors were installed away from paved surfaces, heavy traffic, and mechanical noise sources. Vertical component of noise records was used in the analysis. Records were baseline corrected, and a linear trend correction was applied. The analysis was performed on 4-hour time windows, even for the large aperture radii arrays, to facilitate direct comparison of results. Each

4-hour record was divided into smaller time windows of 200 sec duration with 50% overlap of consecutive windows. Each of these windows, was passed through different narrow frequency band filters (Butterworth, 8 pole, zero-phase). The results were checked against their consistency. For each station pair, SPAC's coefficients were computed using the filtered traces and averaged as a function of frequency for all time windows, common to a given station pair. By inverting the average SPAC's coefficients simultaneously for all distances available in each site, the corresponding phase velocity dispersion curve of the fundamental mode of Rayleigh waves was determined. The curves were inverted, using the iterative inversion code of Herrmann (2004) and the distribution of the Vs values with depth had resulted.

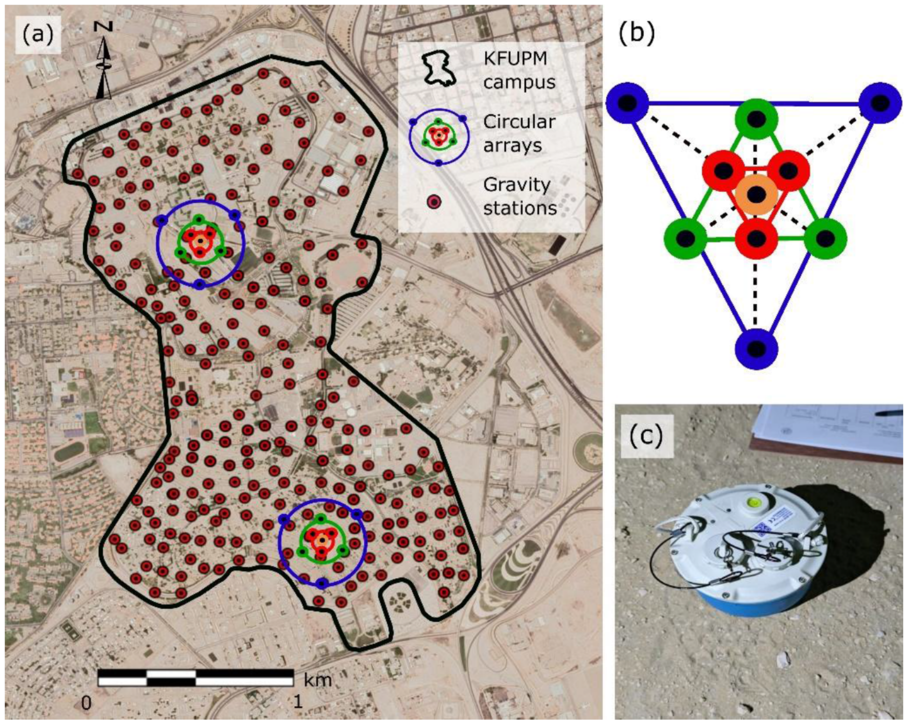

Figure 1—(a) Satellite map of the KFUPM campus in Dhahran, Saudi Arabia, showing the locations of two installed SPAC seismic arrays used for ambient noise analysis and previously collected gravity points. (b) Schematic configuration of the triangular SPAC array setup, highlighting the nested ring layout of sensors with different radii. (c) Photo of a broadband seismic sensor deployed at one of the array stations on the KFUPM campus.

A total of 235 gravity stations were collected across an area of approximately 3.5 km², covering both academic and residential sections of the campus with an approximate station spacing of 100 meters. Data acquisition was carried out exclusively during nighttime hours to minimize the impact of ambient noise caused by daytime vehicular traffic, construction activity, and elevated temperatures. Each gravity station was measured using a Scintrex CG5 gravimeter, which was programmed to perform three consecutive 60-second cycles per station. The gravimeter was set to automatically correct for instrumental tilt, reject spikes, and apply basic tidal corrections during acquisition. To enhance data quality, strict quality control criteria were used in the field. Stations exhibiting tilt errors exceeding 10 arcseconds were discarded and remeasured. This robust protocol resulted in high-precision measurements, with the standard deviation of raw gravity data ranging from 8 to 44 μGal, and an average of 21 μGal across the dataset. Station positioning and elevation were obtained using a Trimble R10+ Differential GPS, providing horizontal and vertical accuracy. The collected gravity data underwent a series of standard corrections, including instrumental drift, earth tide effects, and sensor height adjustments. Drift corrections were modeled using polynomial fitting within the SRGM software environment, which also offered integration with TSOFT (Van Camp and Vauterin, 2005) for accurate tidal modeling. The corrected gravity values were then processed using

Oasis Montaj™ software with the Terrain Correction extension. As no absolute gravity reference station was available in the region, the gravity value at the survey base station was calculated using the theoretical value from the GRS80 geodetic reference system.

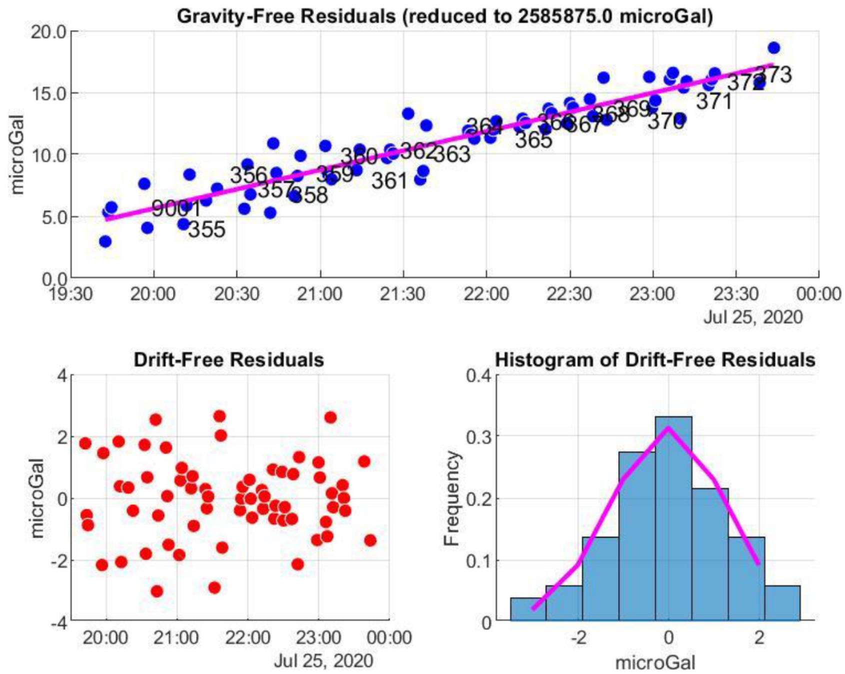

Figure 2—Output from SRGM software illustrating the processing of gravimetric. The upper plot shows raw gravity-free residuals with a polynomial drift correction (magenta line) applied over time. The lower-left scatterplot displays the resulting drift-free residuals, and the lower-right histogram confirms a near-Gaussian distribution of drift-free residuals, validating the quality and stability of the gravity measurements.

Results

The integrated application of SPAC and gravity methods at the KFUPM campus provided complementary insights into the subsurface structure, effectively reducing ambiguity in geophysical interpretations. The results offer a coherent understanding of the sedimentary layering, tectonic deformation, and lateral density variations underlying the campus, consistent with the regional context of the Dammam Dome. The SPAC-derived shear wave velocity (Vs) profiles revealed a distinct multi-layered structure across both survey sites, in the shallowest meters. At greater depths, velocities increased steadily, indicative of transition into more stiff geological formations, likely representing the upper Rus or basal Dammam Formation. These velocity profiles, then converted to density using the empirical relationship of Gardner et al. (1974). These values provided crucial constraints for calibrating the gravity model and mitigating the inherent non-uniqueness of potential field data.

The gravity survey produced a detailed Bouguer anomaly map after applying standard corrections. The complete Bouguer anomaly map revealed distinct lateral density variations across the campus. A dominant gravity high was observed in the northwest quadrant, spatially aligned with outcrops of denser carbonate rocks and correlating with the upper Dammam Formation. In contrast, gravity lows appeared in central and southeastern sectors, suggesting the presence of structurally disrupted or less dense material. These anomalies potentially reflect dissolution features or fault zones, particularly in areas where near-surface sedimentation or karstic development is plausible. Residual gravity anomalies, obtained after removing the regional trend using a first-order polynomial fit, allowed enhanced resolution of local density contrasts. A pronounced residual gravity low was observed trending ENE–WSW across the central part of the campus, coinciding with the previously mapped fracture systems. To better delineate structural boundaries, edge-enhancement filters were applied to the residual gravity data. Horizontal gradient magnitude (HGM) filtering revealed strong lateral density contrasts bordering the central anomaly, indicating subvertical contacts or fault zones. These features coincided spatially with the zones of velocity inversion in the SPAC models and with mapped joint patterns, reinforcing their structural significance. Tilt derivative filtering added further resolution to the anomaly interpretation by amplifying strong signal edges.

Finally, the 3D inversion of the residual Bouguer anomalies, using the VOXI Earth Modeling module, generated volumetric density models of the subsurface. The inversion, constrained by the SPAC-derived density profiles, provided improved depth resolution and reduced model ambiguity. High-density zones clustered beneath the gravity highs are consistent with competent carbonate sequences. Conversely, low-density volumes appeared beneath gravity lows and aligned with SPAC-inferred low-velocity zones, suggesting a high probability of karstification, fractured zones, or detachment surfaces. Together, the SPAC and gravity results point to a complex subsurface structure beneath the KFUPM campus, characterized by lateral heterogeneity, vertical layering, and localized low-density anomalies. The dominant ENE–WSW trend observed in both datasets reflects the influence of regional stress fields associated with the Arabian-Eurasian plate interaction. These structural features diverge from the previously emphasized NNW–SSE joint sets, suggesting a more intricate tectonic regime involving soft-sediment detachments and possible transtensional dynamics, in line with recent findings by Tranos & Osman (2021).

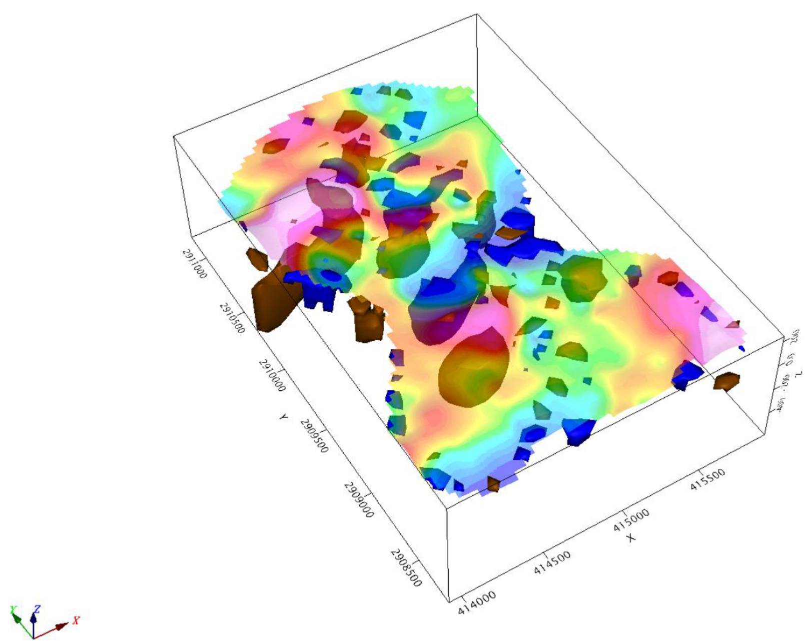

Figure 3—3D VOXI gravity inversion model showing volumetric density distribution beneath the KFUPM campus. It consists of two groups: the blue isosurfaces for bodies with negative density contrast (<-0.08 g/cm³) and the brown isosurfaces for bodies with positive density contrast (>0.05 g/cm³), with their calculated gravity field overlaid on top.

Conclusions

This study presents a successful integration of SPAC analysis and gravity surveying as a novel geophysical workflow for characterizing the urban subsurface. By combining passive seismic data with high-resolution gravity measurements, we demonstrate a synergistic approach that addresses the limitations inherent in either method when applied independently, particularly in environments where logistical and environmental constraints limit the effectiveness of traditional seismic surveys. The SPAC method, based on ambient noise analysis, effectively resolved shear wave velocity (Vs) profiles down to intermediate depths, capturing key transitions between unconsolidated sediments and competent carbonate formations within the Rus and Dammam stratigraphic units. These Vs profiles were translated into density estimates using established empirical relationships, providing critical constraints for refining gravity-based models. The use of ten broadband wireless sensors in concentric circular arrays allowed flexibility in deployment across the KFUPM campus, even in settings challenged by anthropogenic noise and limited open space. The ability to passively record microtremors for several hours and to extract reliable dispersion curves demonstrates the applicability of SPAC in complex urban terrains.

Gravity measurements, acquired through a dense network of 235 stations and corrected for tidal, instrumental, terrain, and drift effects, revealed a heterogeneous density structure beneath the campus. The complete Bouguer anomaly map provided a detailed view of lateral density variations. The integration of SPAC-derived density constraints significantly reduced the non-uniqueness in gravity inversion, improving both vertical resolution and interpretational confidence. The combined geophysical data indicate a complex subsurface characterized by alternating high- and low-density zones, vertical layering, and a dominant ENE–WSW structural trend. This orientation diverges from the historically mapped NNW–SSE joint patterns and aligns instead with the regional maximum horizontal stress direction (σ_H), associated with the Arabian-Eurasian plate collision. The observation of low-density, low-velocity anomalies in central and southeastern portions of the campus is consistent with features interpreted as karstic dissolution zones, soft-sediment detachments, or fault-fracture corridors. The 3D inversion of residual gravity anomalies using the VOXI platform allowed further structural interpretation of the subsurface.

Importantly, this research underscores the feasibility and advantages of integrating passive seismic and gravity methods in urban environments. The low-impact nature of both approaches, particularly when conducted outside of peak activity hours, makes them ideal for applications in university campuses, heritage zones, and densely built-up areas where conventional active surveys may be impractical or undesirable. The successful deployment and interpretation of the joint dataset highlights the potential for this methodology to be extended to other regions with similar challenges. In conclusion, the combined SPAC–gravity approach applied at the KFUPM campus has provided valuable insights into the subsurface architecture of the Dammam Dome. It contributes to a deeper understanding of local tectonic patterns, sedimentary structures, and possible geohazard features. This integrated methodology provides a scalable and non-invasive framework for future urban geophysical surveys, supporting enhanced geotechnical, environmental, and seismic hazard assessments across a wide range of disciplines.

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References

1. Aki, K. (1957). Space and Time Spectra of Stationary Stochastic Waves, with Special Reference to Microtremors, *Bulletin of the Earthquake Research Institute*, **25**, 415–457.
2. Brocher T.M. (2005) Empirical Relations between Elastic Wavespeeds and Density in the Earth's Crust. *Bulletin of the Seismological Society of America*, **95** (6), 2081–2092. https://doi.org/10.1785/0120050077
3. Chavanidis K., Stampolidis A., Kirmizakis P., Tranos M., Fedi M., Pasteka R., Al-Ramadan K., Kaka S., Tsokas G., & Soupios P. (2022). Gravity Survey on the Oil-Bearing Dammam Dome (Eastern Saudi Arabia) and Its Implications. *Remote Sens*., **14**, 735. https://doi.org/10.3390/rs14030735
4. Cheng F., Xia J., Xu Y., Xu Z., & Pan Y. (2015). A new passive seismic method based on seismic interferometry and multichannel analysis of surface waves. *Journal of Applied Geophysics*, **117**, 126–135. https://doi.org/10.1016/j.jappgeo.2015.04.005

5. Cho I., & Iwata T. (2020). Limits and Benefits of the Spatial Autocorrelation Microtremor Array Method Due to the Incoherent Noise, With Special Reference to the Analysis of Long Wavelength Ranges. *Journal of Geophysical Research: Solid Earth*, **126** (2). https://doi.org/10.1029/2020JB019850
6. Gabàs, A., Macau A., Benjumea, B., Bellmunt F., Figueras S., & Vilà M. (2014). Combination of Geophysical Methods to Support Urban Geological Mapping. *Surv Geophys* 2014, **35**, 983–1002. https://doi.org/10.1007/s10712-013-9248-9
7. Gardner G.H., Gardner L.W., Gregory A.R. (1974). Formation velocity and density - The diagnostic basics for stratigraphic traps. *Geophysics*, **39** (6), 770–780. https://doi.org/10.1190/1.1440465
8. Herrmann, R. (2004). *Computer programs in seismology*. Version 3.30, Saint Louis University, Missouri.
9. Luo S., Luo Y., Zhu L., & Xu Y. (2016). On the reliability and limitations of the SPAC method with directional wavefield. *Journal of Applied Geophysics*, **126**, 172–182. https://doi.org/10.1016/j.jappgeo.2016.01.023
10. Morgan D., Gunn D., Payo A., & Raines M. (2022). Passive Seismic Surveys for Beach Thickness Evaluation at Different England (UK) Sites. *Journal of Marine Science and Engineering*, **10** (5), 667. https://doi.org/10.3390/jmse10050667
11. Soupios P., Stampolidis A., Fedi M., Kaka S., Al-Ramadan K., Tsokas G., & Pasteka R. (2021). Gravity Survey of King Fahd University of Petroleum and Minerals Dammam Dome, Saudi Arabia. *SPE Middle East Oil & Gas Show Conference and Conference*. https://doi.org/10.2118/204564-MS
12. Tranos M.D., & Osman M.S. (2021). Rus detachment in Dammam Dome, Eastern Saudi Arabia: A new soft-sediment structure as a “sensitive stress tensor” for the Zagros collision. *Geol. Mag.*, 1–13. https://doi.org/10.1017/S0016756821001217
13. Van Camp M., Vauterin, P. (2005). Tsoft: Graphical and interactive software for the analysis of time series and Earth tides. *Comput. Geosci.* **31**, 631–640. https://doi.org/10.1016/j.cageo.2004.11.015
14. Weijermars, R. (1999). Surface geology, lithostratigraphy and tertiary growth of the Dammam Dome, Saudi Arabia: A new field guide. *GeoArabia*, **4**, 199–226. https://doi.org/10.2113/geoarabia0402199